

A
Project Report
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1. INTRODUCTION

Fitness, travel, and lifestyle management have become an integral part of modern living. In recent years, customer demand for online travel booking platforms has surged significantly. Travelers today seek hassle-free ways to discover tour destinations, compare travel packages, inspect inclusions, check departure dates, and book vacations online without needing physical travel agency visits.

The 'AtlasGo - Online Travel & Tour Booking System' is a dynamic web application developed to allow users to browse, search, filter, and book domestic and international tour packages online. Built using PHP for server-side scripting and MySQL for relational database management, AtlasGo enables customers to explore handpicked destinations, select tour dates, input guest numbers, choose payment options, and track their booking statuses.

The system is divided into two major parts:

- **Client Side (User Portal):** Where customers browse destinations, filter tour packages by price or region, submit online booking requests, view photo galleries, and track their reservation status under 'My Bookings'.
- **Admin Side (Management Panel):** Where the administrator manages the entire catalog of destinations, tour packages, photo gallery showcase, registered user accounts, and customer booking approvals.

Both sides interact with a unified MySQL database ('project'), ensuring that any modification executed from the admin panel—such as adding a new tour package or updating destination images—is instantly reflected on the customer-facing website.

1.1 Problem Statement

Traditional paper-based or offline travel booking mechanisms present several challenges for both customers and tour operators:

- **Manual Booking Friction:** Customers must visit physical offices or make repeated phone calls to check package availability, pricing, and dates.
- **Lack of Package Transparency:** Detailed day-wise itineraries, inclusion lists, and high-resolution destination photos are rarely accessible in one place.
- **Inconvenient Reservation Tracking:** Travelers lack a centralized digital dashboard to verify whether their booking status is 'Pending', 'Processing', or 'Completed'.
- **Administrative Overhead:** Operators struggle to maintain paper registers, manage package price revisions, and track customer details manually.

1.2 Purpose & Scope

The scope of AtlasGo encompasses a complete web-based solution for tour package discovery and reservation management. Key functional boundaries include:

- Dynamic filtering of tour packages across Domestic (Kashmir, Goa, Kerala, Jaipur, Leh, Udaipur, Andaman) and International (Dubai, Maldives, US, Switzerland, Japan, Thailand, Vietnam, Bali, France) regions.
- User authentication and session state maintenance (`tbl\_user`).
- Booking creation workflow with payment mode selection (Net Banking, UPI, Cash) and status tracking (`tbl\_booking`).
- Admin dashboard empowering full CRUD operations on Destinations, Packages, Gallery photos, and Booking approvals.

2. TECHNOLOGY USED

The development of AtlasGo leverages standard web development technologies to ensure performance, security, responsiveness, and ease of maintenance:

2.1 Client-Side Technologies (Front-End)

- HTML5: Structured markup defining pages, forms, tables, and media elements.
- CSS3 & Bootstrap 5: Responsive layout grid, modern typography, glassmorphism cards, modal overlays, and mobile navigation drawer styling.
- JavaScript (ES6+): Client-side form validations, dynamic filter interactions, slider swipers, and mobile menu toggling.
- FontAwesome 7 & Bootstrap Icons: High-quality vector icon sets used across navigation bars, buttons, and summary widgets.

2.2 Server-Side Technologies (Back-End)

- PHP 8.2: Primary server-side language handling procedural/object-oriented logic, user authentication sessions, image file uploads, dynamic SQL query composition, and page rendering.
- MySQLi Extension: Secure PHP database interface for executing parameterized queries and handling database recordsets.

2.3 Database Management System

- MySQL / MariaDB 10.4: Relational database engine storing user accounts, destination categories, package details, bookings, and gallery paths.
- phpMyAdmin: Web GUI interface used for schema creation, table structure setup, foreign key constraints, and SQL data dumps.

2.4 Development & Hosting Environment

- XAMPP Stack: Apache 2.4 web server and MariaDB server environment.
- Visual Studio Code: Primary code editor equipped with syntax highlighting, debugging tools, and Git integration.

3. SYSTEM REQUIREMENTS

3.1 Hardware Requirements

Recommended hardware specifications for hosting and running AtlasGo:

- 1) Developer / Server Machine:
 - Processor: Intel Core i3 / i5 or AMD Ryzen 3 / 5 (2.0 GHz or faster)
 - RAM: 8 GB DDR4 minimum
 - Storage: 256 GB SSD with 5 GB free disk space
 - Network: High-speed Wi-Fi / Broadband connection
- 2) Client / End-User Machine:
 - Device: Laptop, Desktop, Tablet, or Smartphone
 - RAM: 2 GB minimum
 - Display Resolution: 1024 x 768 or higher

3.2 Software Requirements

- Operating System: Windows 10/11, macOS, or Linux
- Web Server: Apache 2.4 (via XAMPP)
- PHP Engine: PHP version 8.0 or higher
- Database Engine: MariaDB 10.4+ or MySQL 5.7+
- Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari

4. OBJECTIVES OF THE SYSTEM

The core objectives of the AtlasGo project are:

- To design an intuitive, responsive website for browsing and booking travel packages online.
- To implement complete CRUD (Create, Read, Update, Delete) operations for Destination, Package, Gallery, User, and Booking modules.
- To handle file uploads and image management—saving uploaded package thumbnails, destination banners, and gallery photos into the server uploads directory.
- To provide price-based and region-based search and filtering tools so travelers can quickly find their desired vacation packages.
- To implement a streamlined booking reservation workflow allowing registered users to select dates, specify passenger details, select payment modes, and confirm bookings.
- To maintain relational integrity across database tables (`tbl\_user`, `tbl\_destination`, `tbl\_package`, `tbl\_booking`, `tbl\_gallery`) using MySQL foreign keys.
- To build an administrative control dashboard for monitoring total revenue metrics, package management, user directories, and updating customer booking statuses.
- To allow customers to track their booking reservation history and live payment status ('Pending', 'Processing', 'Completed') through a dedicated 'My Bookings' page.

5. SYSTEM ANALYSIS & MODELING

System analysis examines the problem domain to establish functional requirements, while system design structures the technical solution. Both Data Flow Diagrams (DFD) and Unified Modeling Language (UML) diagrams were designed to represent data transformations, static entity structures, and dynamic system interactions.

5.1 System Flowchart

The flowchart below illustrates the end-to-end operational path for visitors, registered users, and administrators:

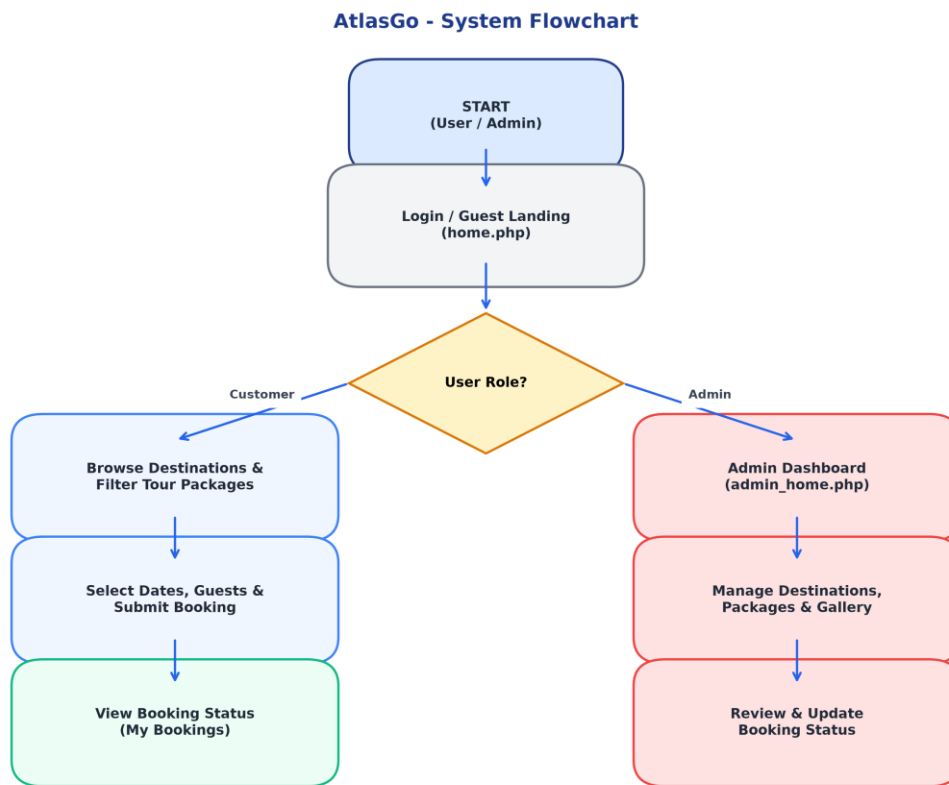


Figure 5.1: AtlasGo End-to-End System Flowchart

5.2 Data Flow Diagrams (DFD)

Data Flow Diagrams visualize data movement, processes, and data stores.

Context Level DFD (Level 0 DFD)

Presents the entire AtlasGo system as a single central process interacting with Customer and Admin external entities:

Context Level Data Flow Diagram (Level 0 DFD)

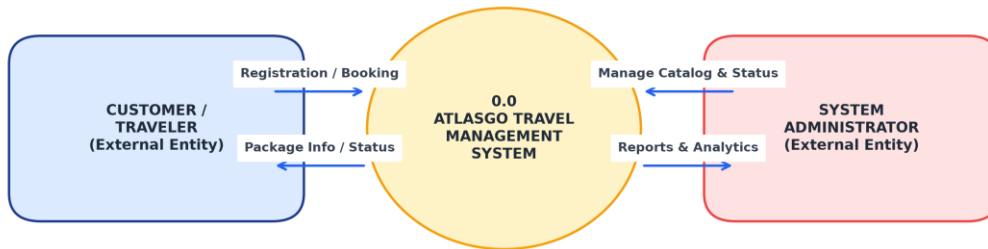


Figure 5.2: Context Level Data Flow Diagram (Level 0 DFD)

Level 1 DFD

Decomposes the system into major functional processes (1.0 Auth, 2.0 Destinations, 3.0 Packages, 4.0 Bookings) and their data flow connections to database stores:

Level 1 Data Flow Diagram (DFD)

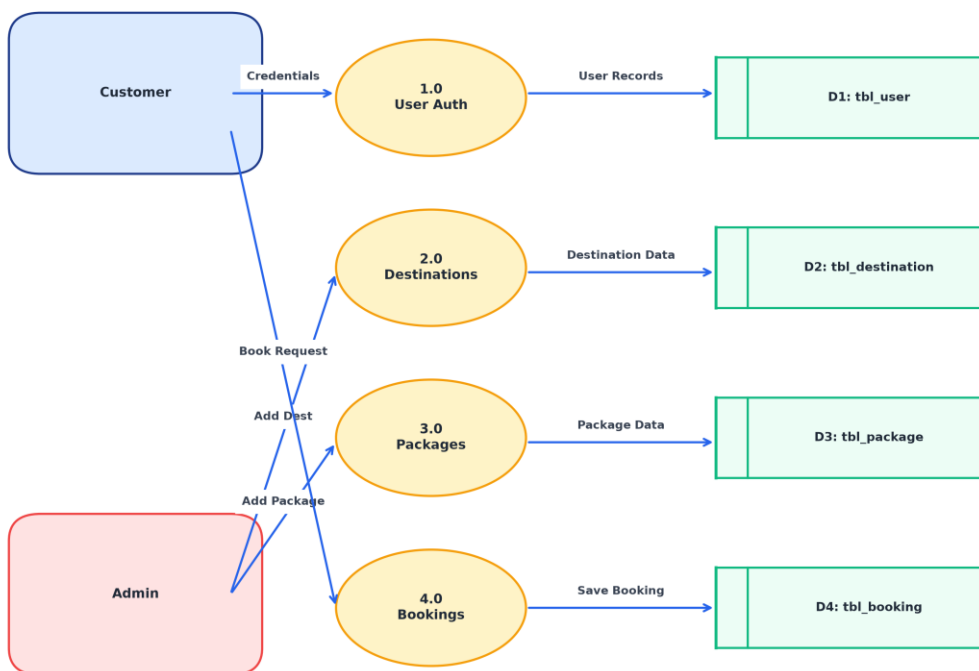


Figure 5.3: Level 1 Data Flow Diagram (DFD)

Level 2 DFD (Process 4.0 - Booking Processing)

Details sub-processes involved in package validation, recording booking details, and status updates by the administrator:

Level 2 DFD - Process 4.0 (Booking Processing)

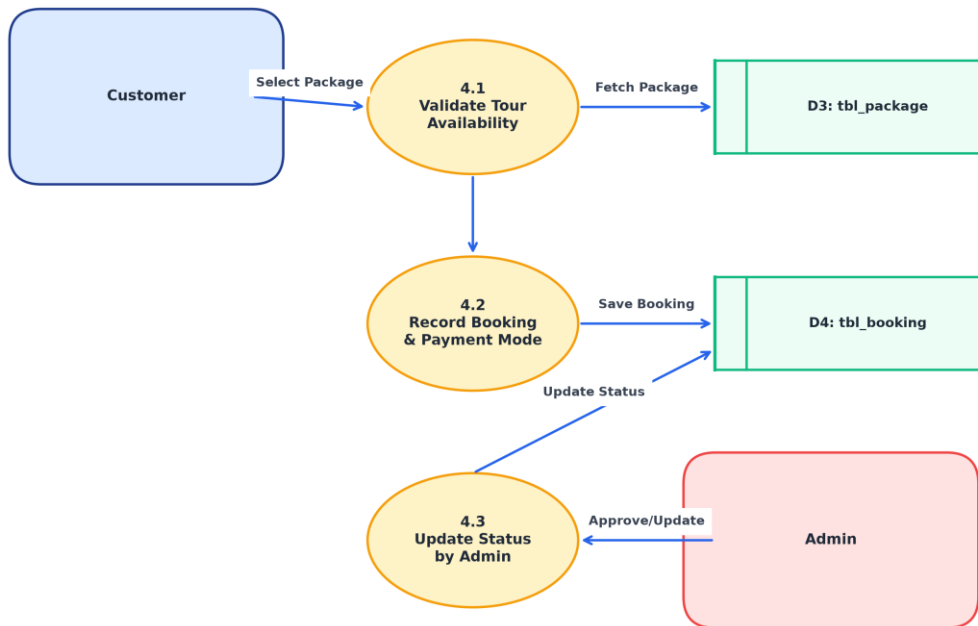


Figure 5.4: Level 2 Data Flow Diagram for Booking Sub-processes

5.3 Unified Modeling Language (UML) Diagrams

UML provides standardized graphical diagrams for object-oriented modeling.

Use Case Diagram

Defines functional system services from the perspective of Customer and Admin actors:

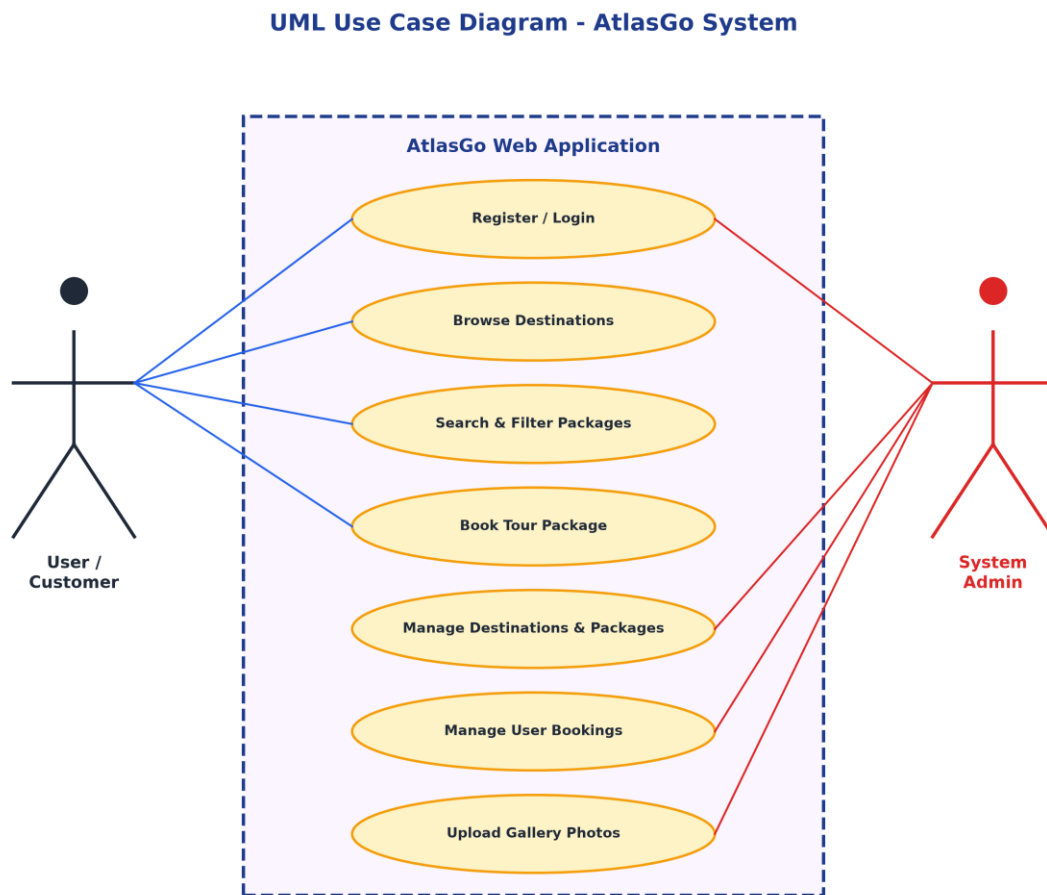


Figure 5.5: UML Use Case Diagram for User and Admin Roles

Class Diagram

Represents static class structures, attributes, methods, and relational multiplicities:

UML Class Diagram - AtlasGo Domain Model

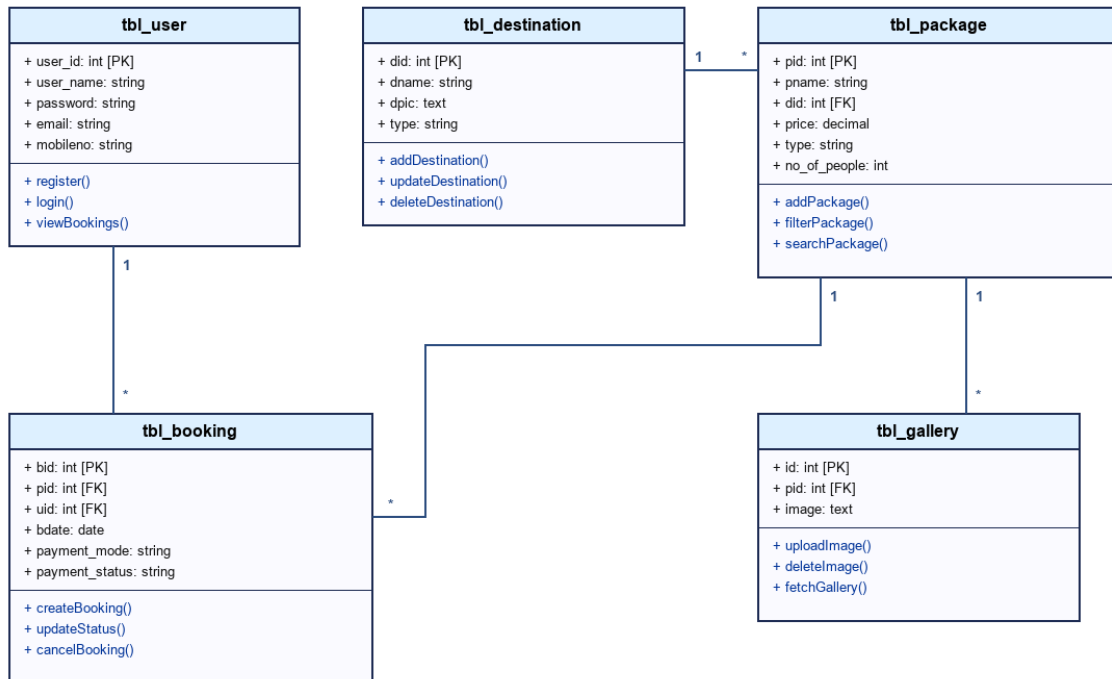


Figure 5.6: UML Class Diagram for AtlasGo Domain Model

Activity Diagram :

- Represents the step-by-step workflow for customer tour browsing, authentication checking, form submission, and record creation:

UML Activity Diagram - User Tour Booking Workflow

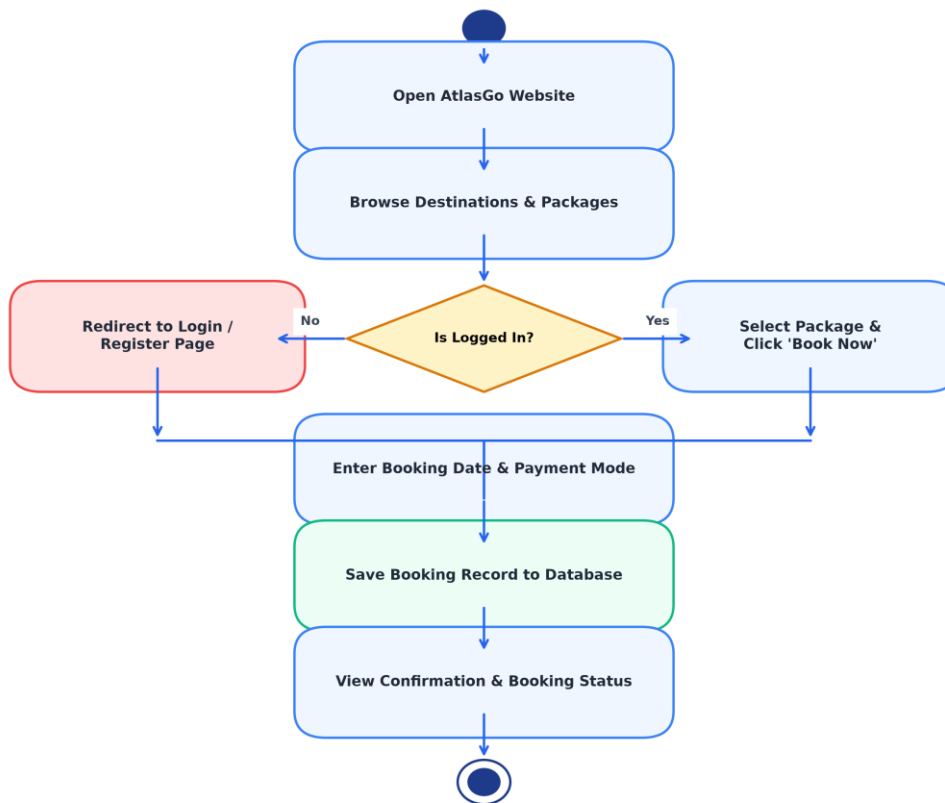


Figure 5.8: UML Activity Diagram for Customer Tour Reservation Workflow

6. DATABASE DESIGN

The database for AtlasGo is named `project`. It contains five primary relational tables designed to store user accounts, destination categories, tour packages, booking requests, and photo gallery paths.

6.1 Data Dictionary & Table Definitions

1. Table: `tbl\_user`

Stores registered customer credentials and contact info.

Field Name	Data Type	Constraint	Null	Description
user_id	INT(11)	PRIMARY KEY, AUTO_INC	No	Unique User ID
user_name	VARCHAR(50)	None	No	Customer Username / Full Name
password	VARCHAR(50)	None	No	User Login Password
email	VARCHAR(100)	None	No	Customer Email Address
mobilen0	VARCHAR(10)	None	No	10-digit Contact Number

2. Table: `tbl\_destination`

Stores destination categories and banner image paths.

Field Name	Data Type	Constraint	Null	Description
did	INT(11)	PRIMARY KEY, AUTO_INC	No	Unique Destination ID
dname	VARCHAR(80)	None	No	Destination Name (e.g. Kashmir)
dpic	TEXT	None	No	Uploaded Banner Image Path
type	VARCHAR(100)	None	No	Domestic or International

3. Table: `tbl\_package`

Stores tour package details, pricing, duration, and foreign key reference to `tbl\_destination`.

Field Name	Data Type	Constraint	Null	Description
pid	INT(11)	PRIMARY KEY, AUTO_INC	No	Unique Package ID
pname	VARCHAR(100)	None	No	Tour Package Title
did	INT(11)	FOREIGN KEY	No	References tbl_destination(did)
pic	TEXT	None	No	Package Thumbnail Image Path

type	VARCHAR(80)	None	No	Domestic / International
price	DECIMAL(10,0)	None	No	Package Price (INR)
description	TEXT	None	No	Detailed Tour Itinerary
no_of_people	INT(11)	None	No	Max Capacity / Travelers
date	DATE	None	No	Departure Date
time	VARCHAR(50)	None	No	Departure Time

4. Table: `tbl\_booking`

Stores customer booking transactions and payment status.

Field Name	Data Type	Constraint	Null	Description
bid	INT(11)	PRIMARY KEY, AUTO_INC	No	Unique Booking ID
pid	INT(11)	FOREIGN KEY	No	References tbl_package(pid)
uid	INT(11)	FOREIGN KEY	No	References tbl_user(user_id)
bdate	DATE	None	No	Selected Tour Date
payment_mode	VARCHAR(50)	None	No	Net Banking / UPI
payment_status	VARCHAR(80)	None	No	Pending / Processing / Completed
no_of_kids	INT(11)	None	No	Number of Accompanied Children

5. Table: `tbl\_gallery`

Stores showcase gallery image references.

Field Name	Data Type	Constraint	Null	Description
id	INT(11)	PRIMARY KEY, AUTO_INC	No	Unique Image ID
image	TEXT	None	No	Gallery Image File Path

6.2 Table Relationships & Foreign Keys

- `tbl\_package.did` → `tbl\_destination.did`: Connects every package to a specific destination category.
- `tbl\_booking.pid` → `tbl\_package.pid`: Associates a booking reservation with a specific tour package.
- `tbl\_booking.uid` → `tbl\_user.user\_id`: Identifies which customer placed the booking reservation.

7. MODULES DESCRIPTION

7.1 Client-Side Modules

- 1. Home Module (`home.php`): Displays hero banners, category quick links, featured packages, statistics counters, and customer testimonials.
- 2. Destination Explorer (`destination.php`, `destination\_show.php`): Allows travelers to view destinations classified into Domestic (Kashmir, Goa, Kerala, Jaipur, Leh, Udaipur, Andaman) and International (Dubai, Maldives, US, Switzerland, Japan, Thailand, Vietnam, Bali, France).
- 3. Tour Packages & Search (`packages.php`, `package\_filter.php`, `package\_search.php`): Provides package cards with filter controls by price range, destination, and passenger count.
- 4. Package Details Module (`package\_info.php`): Renders complete itinerary descriptions, inclusions, timing, pricing, and a 'Book Now' trigger.
- 5. Booking Module (`addbooking.php`): Handles booking submissions, verifying session authentication, travel dates, child count, and payment mode.
- 6. My Bookings Module (`booking.php`): Displays customer reservation history in a clear tabular format, highlighting payment status badges.
- 7. Gallery Module (`gallery.php`): Displays high-resolution destination photo grids.
- 8. Authentication Module (`login.php`, `register.php`, `logout.php`): Manages user registration, session logins, password checks, and logout redirects.

7.2 Admin-Side Modules

- 1. Admin Dashboard (`admin\_home.php`): Presents administrative summary metrics—total packages, registered users, destinations, active bookings, and gallery count.
- 2. Destination Management (`admin\_add\_destination.php`, `admin\_show\_destination.php`, `admin\_update\_destination.php`): Complete CRUD operations for destinations including thumbnail image uploads.
- 3. Package Management (`admin\_add\_package.php`, `admin\_show\_package.php`, `admin\_update\_package.php`, `admin\_delete\_package.php`): Full CRUD control over tour package details, pricing, itineraries, and media uploads.
- 4. Booking Management (`admin\_show\_booking.php`, `admin\_update\_booking.php`): Review submitted bookings, change payment status (Pending → Completed / Processing), and delete invalid orders.
- 5. User Management (`admin\_show\_user.php`): Displays all registered customer profiles, emails, and contact numbers.
- 6. Gallery Management (`admin\_add\_gallery.php`, `admin\_show\_gallery.php`, `admin\_delete\_gallery.php`): Upload new travel showcase pictures or delete obsolete photos.

8. SYSTEM TESTING

Testing verifies that all modules function as expected, input forms validate data correctly, database queries execute without errors, and administrative CRUD operations work reliably.

Test ID	Module	Test Scenario	Input Data	Expected Result	Status
TC01	User Auth	Register with valid data	Name, Email, Pass, Mobile	Account created & redirect to Login	PASS
TC02	User Auth	Login with wrong pass	User: arman, Pass: 999	Error alert 'Invalid Credentials'	PASS
TC03	Packages	Filter by price range	Select 'Under ₹25,000'	Displays only packages <= ₹25k	PASS
TC04	Booking	Book package when logged in	Date: 2027-01-22, NetBanking	Record saved in tbl_booking	PASS
TC05	Booking	Book package when guest	Click 'Book Now' without login	Redirects to login.php page	PASS
TC06	Admin Dest	Add new Destination	Name: BALI, Pic: bali.png	Destination added to tbl_destination	PASS
TC07	Admin Pkg	Delete package record	Click Delete on PID=25	Record removed from tbl_package	PASS
TC08	Admin Book	Update payment status	Change BID 9 to 'Completed'	Status updated in DB & My Bookings	PASS
TC09	Gallery	Upload new image	File: sunset.jpg	Image saved in uploads & database	PASS
TC10	Session	Logout action	Click Logout link	Session destroyed, redirect to Login	PASS

9. SYSTEM SCREENSHOTS

This section presents visual interface captures of the completed AtlasGo platform, including both customer-facing client pages and administrative management modules.

9.1 User (Client) Side Screenshots

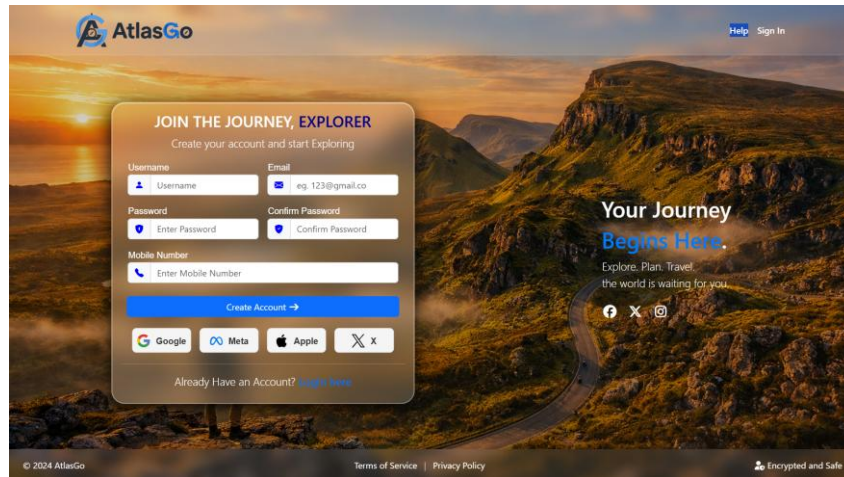


Figure 9.1: Registration Page

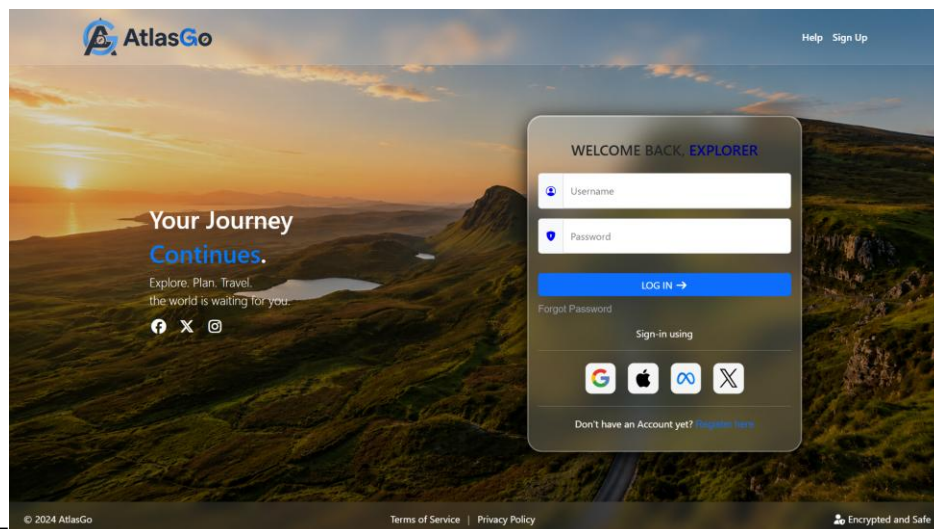


Figure 9.2: Login Page

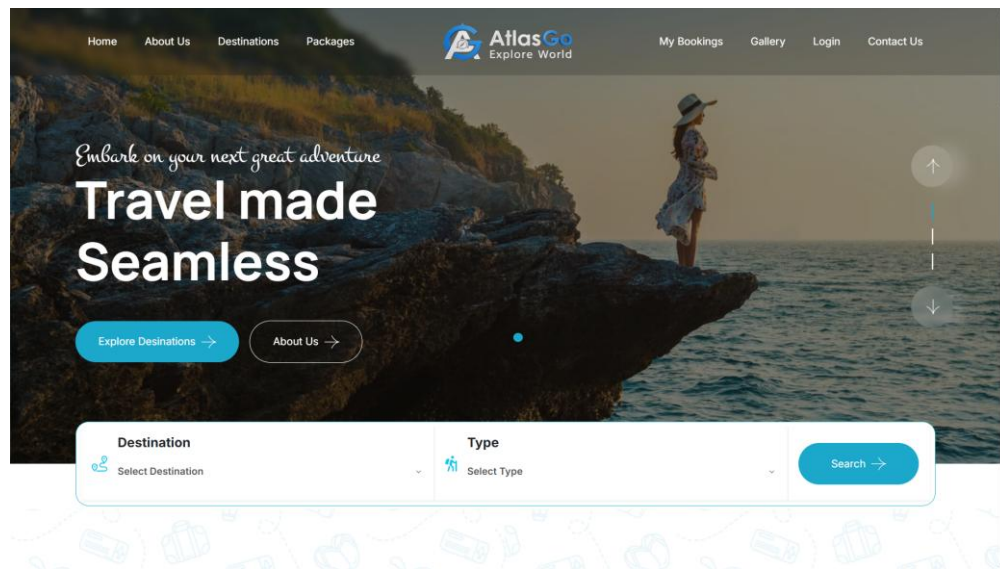


Figure 9.3: Home Page Hero Section & Navigation Bar

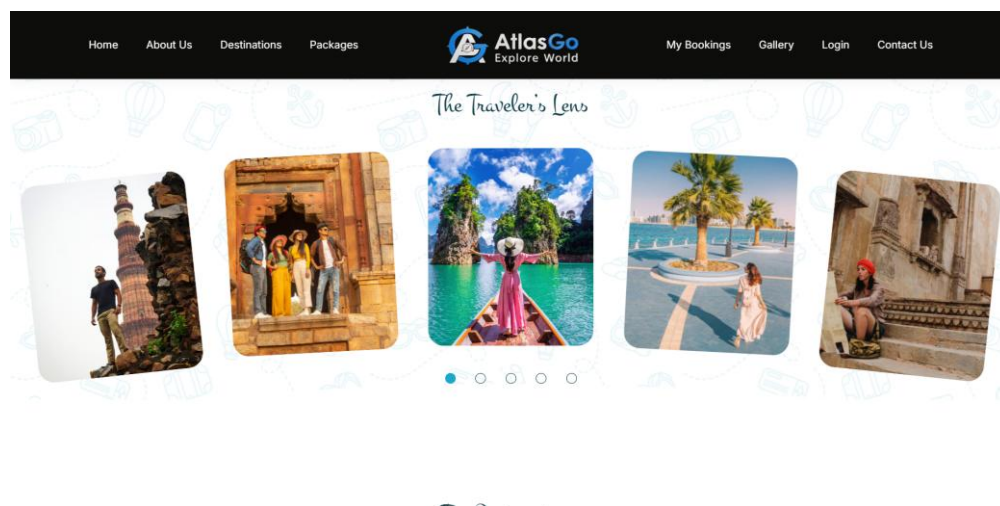


Figure 9.4: Home Page Popular Tour Destinations Showcase

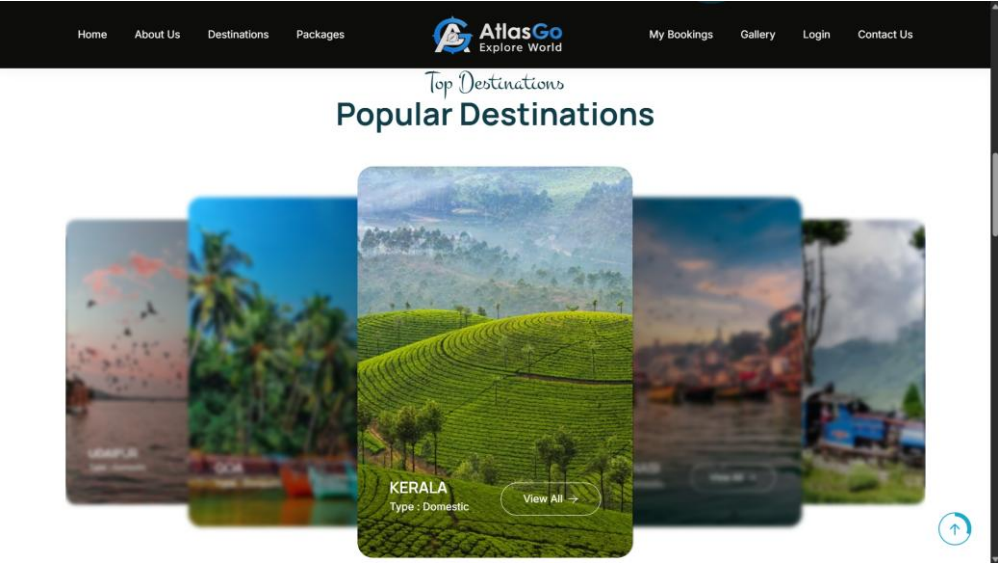


Figure 9.5: Home Page Featured Packages & Special Offers



Figure 9.6: Destinations Overview Page

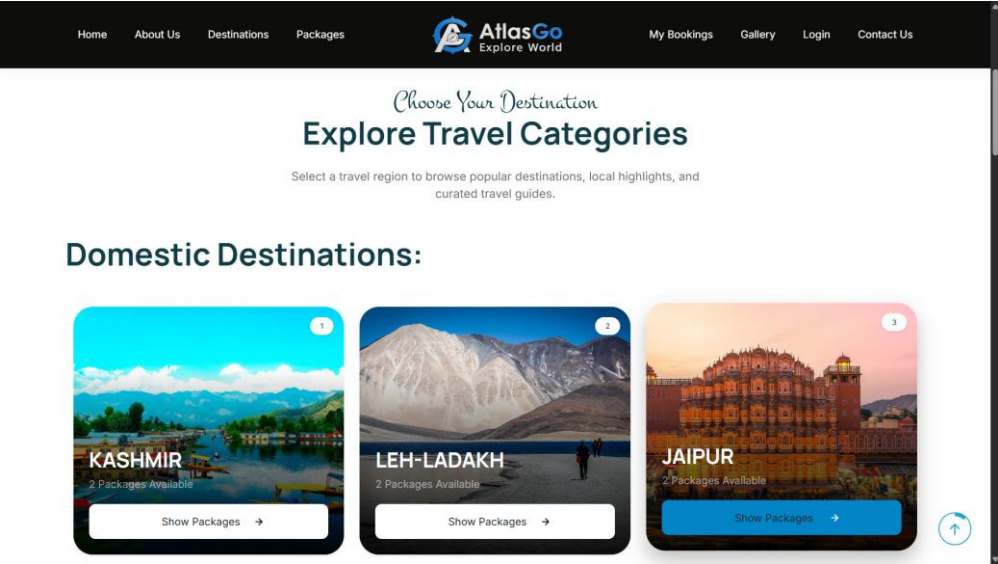


Figure 9.7: Domestic Destinations Category View (Kashmir, Goa, Kerala)

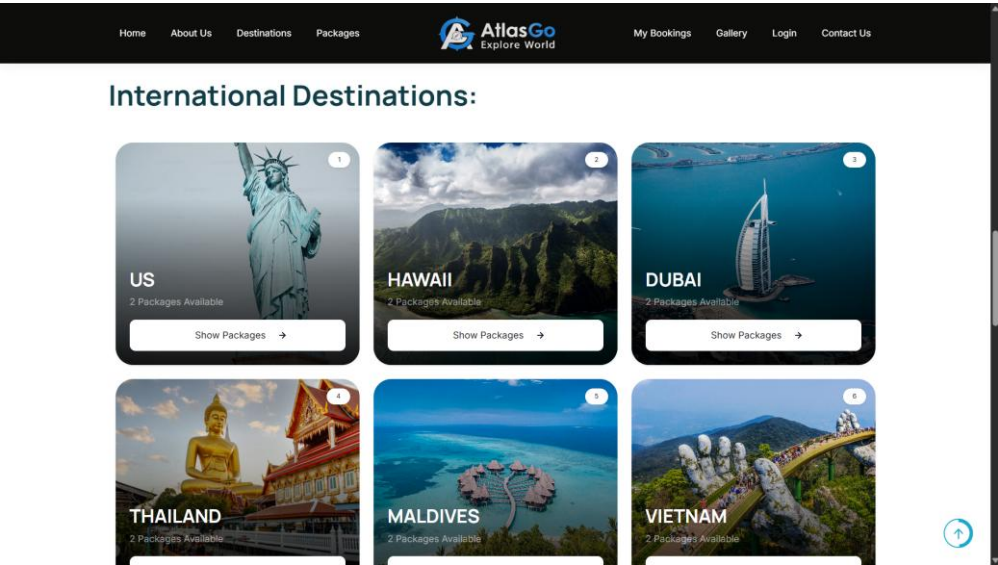


Figure 9.8: International Destinations Category View (Dubai, Maldives, US)

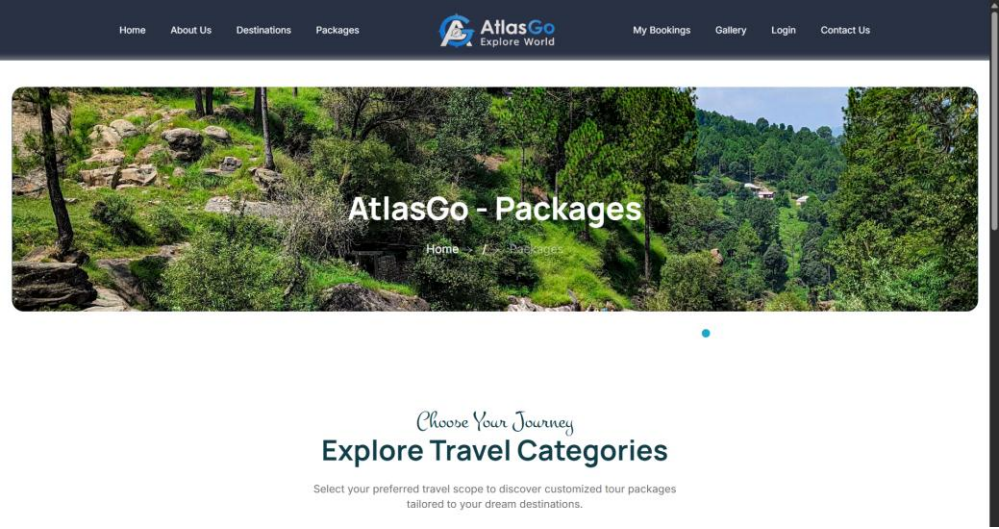


Figure 9.9: Tour Packages Grid & Price Filter Controls

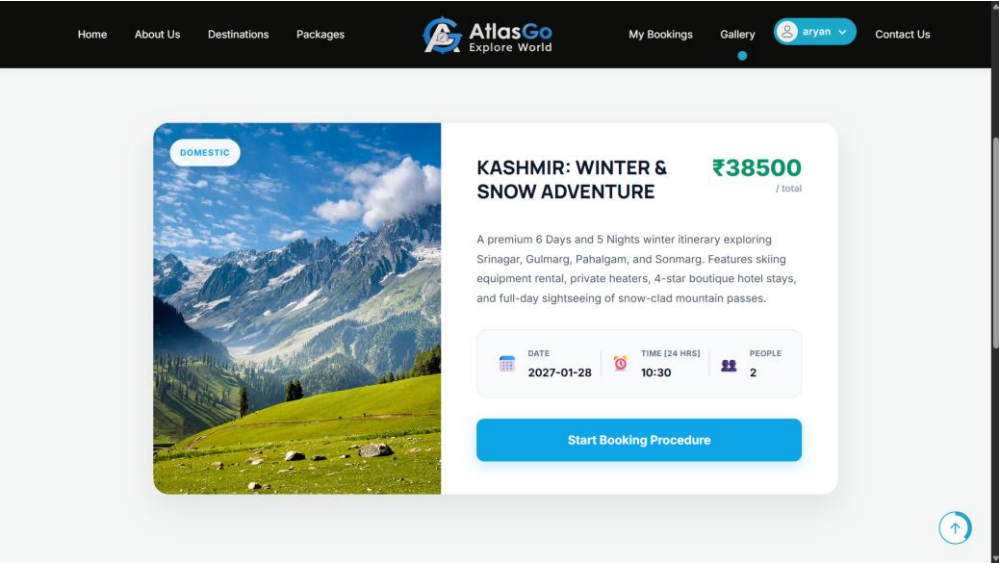


Figure 9.10: Detailed Package Itinerary & Inclusions Page

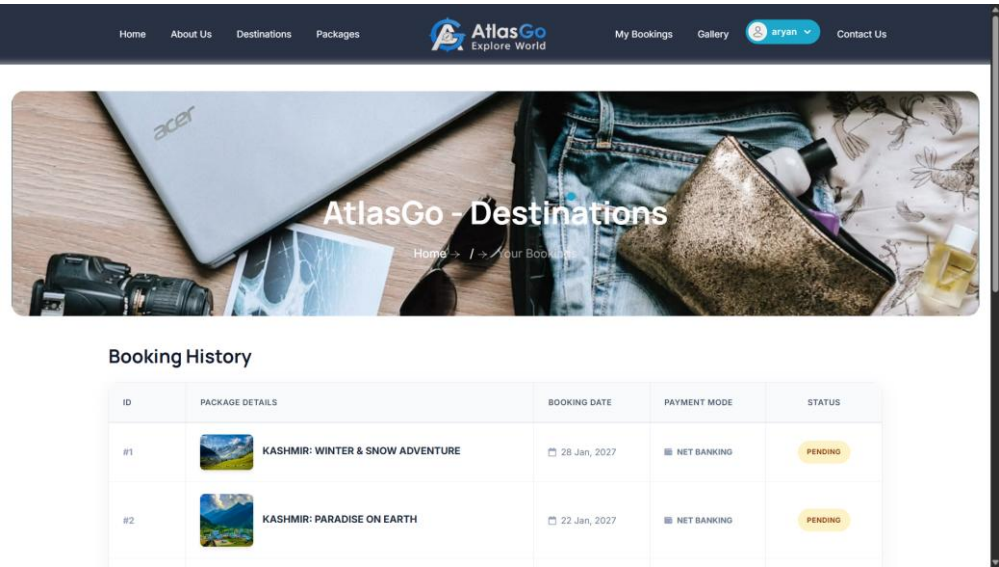


Figure 9.11: Package Booking Request Submission Form

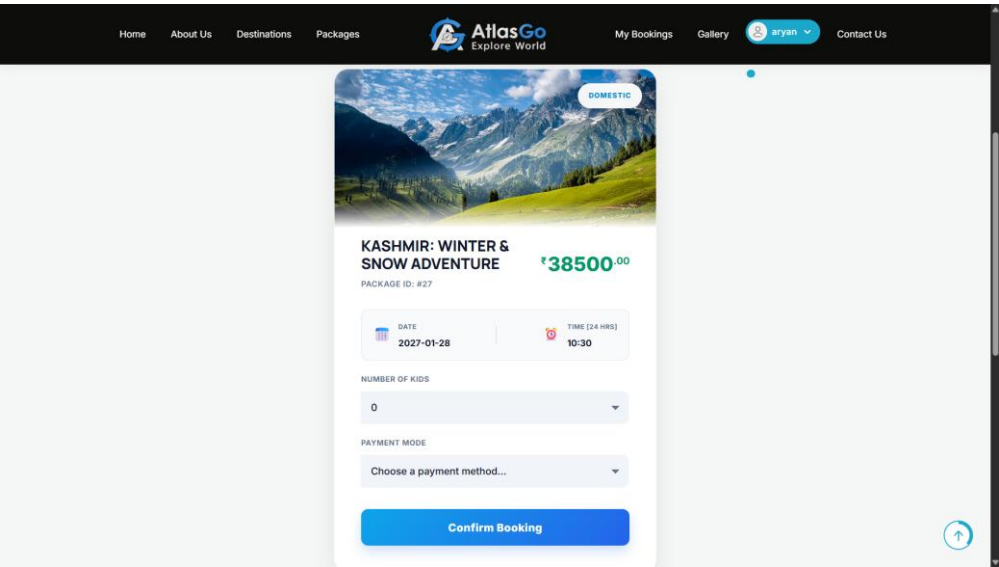


Figure 9.12: Customer 'My Bookings' Status Tracking Table

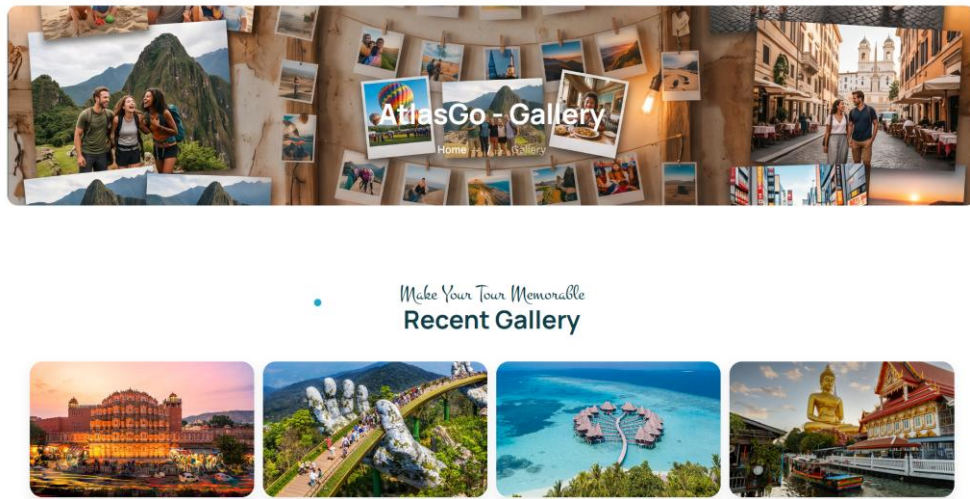


Figure 9.13: High-Resolution Travel Showcase Gallery

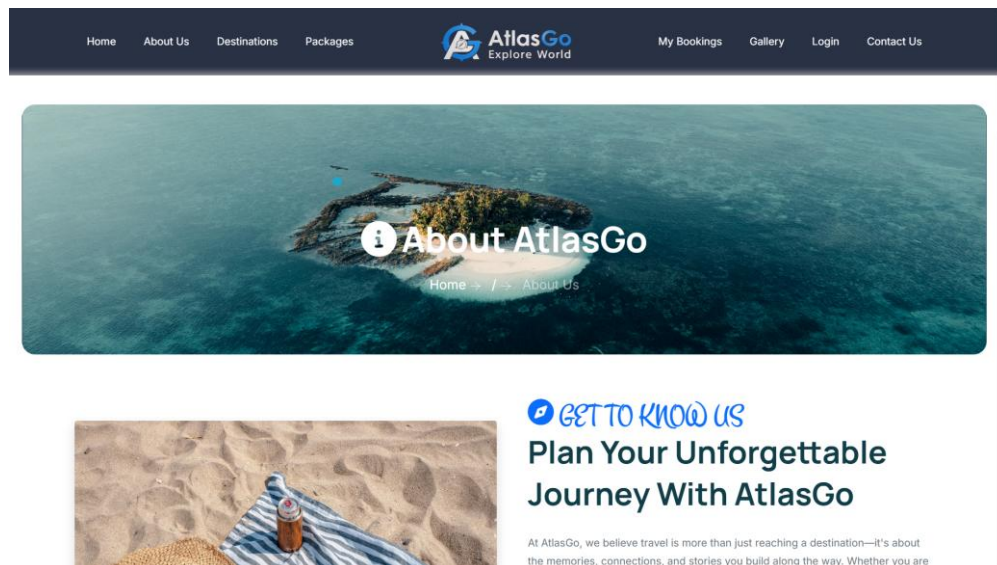


Figure 9.14: About Us Page - Company Overview & Services

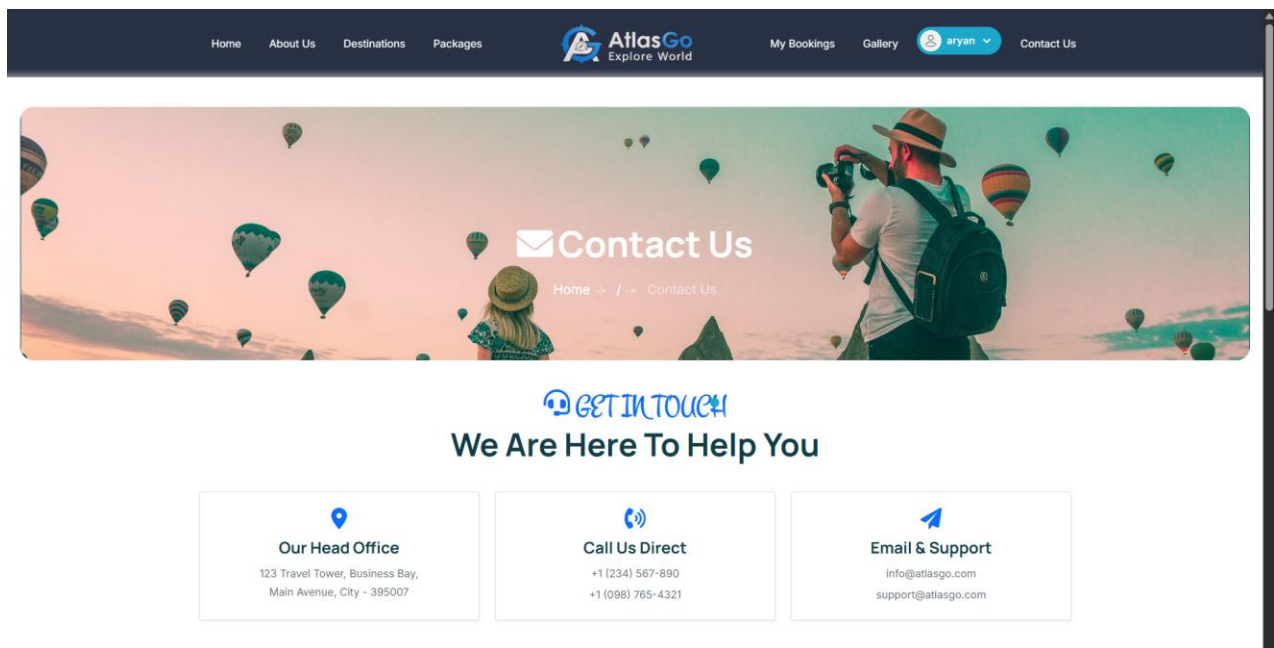


Figure 9.15: Contact Us Page & Customer Query Form

9.2 Admin Side Screenshots

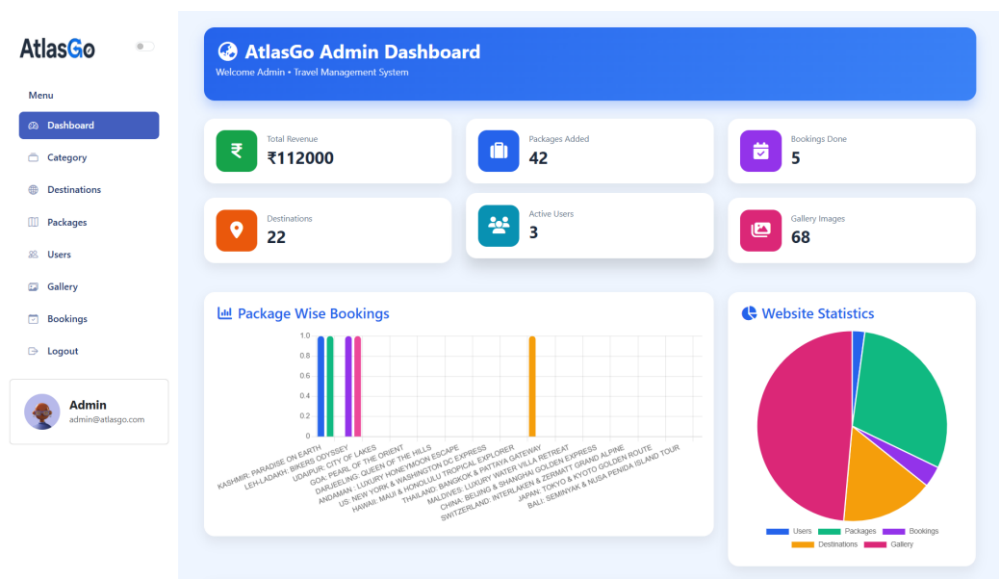


Figure 9.2.1: Admin Dashboard - System Statistics & Summary Widgets

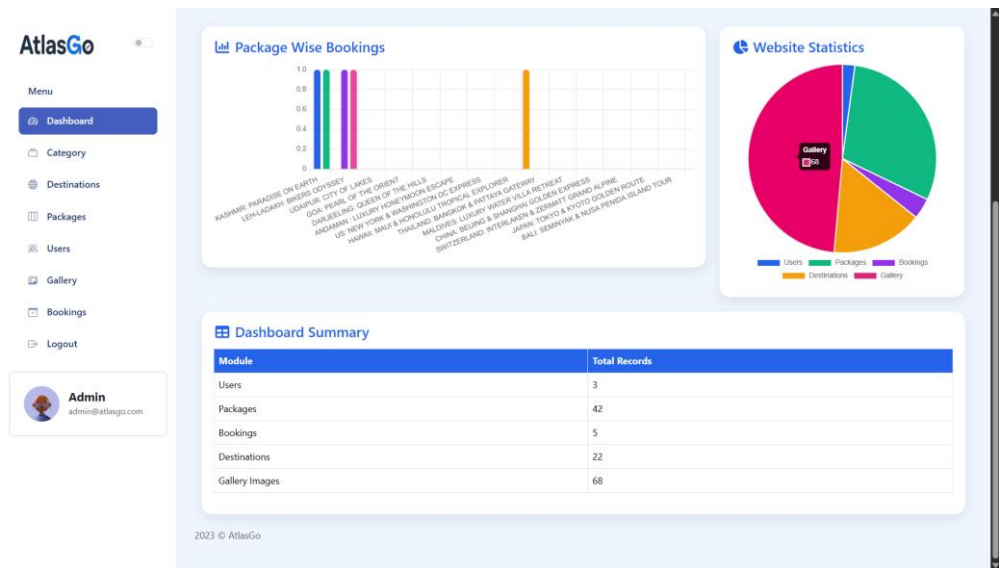


Figure 9.2.2: Admin Dashboard - Quick Management Navigation

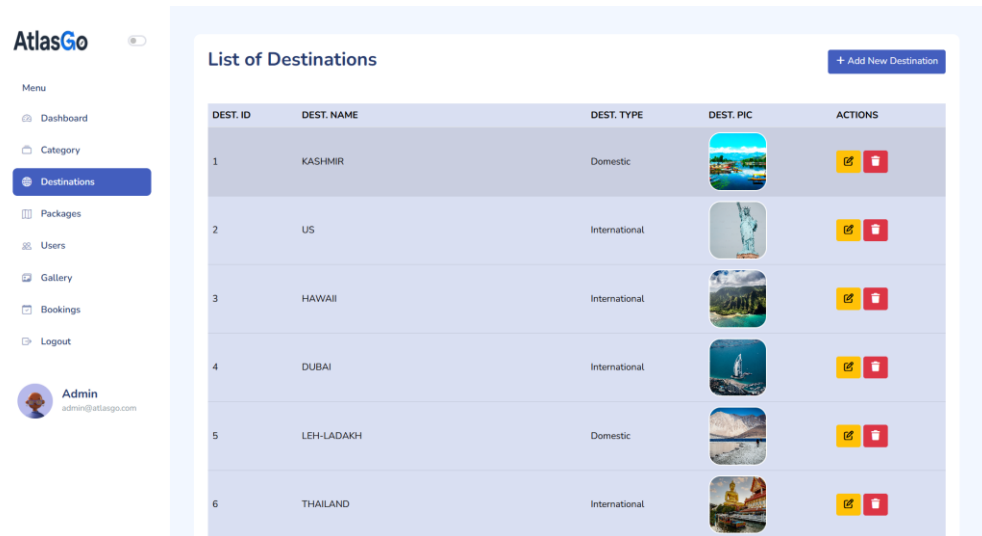


Figure 9.2.3: Admin Destination Management List (Show Destinations)

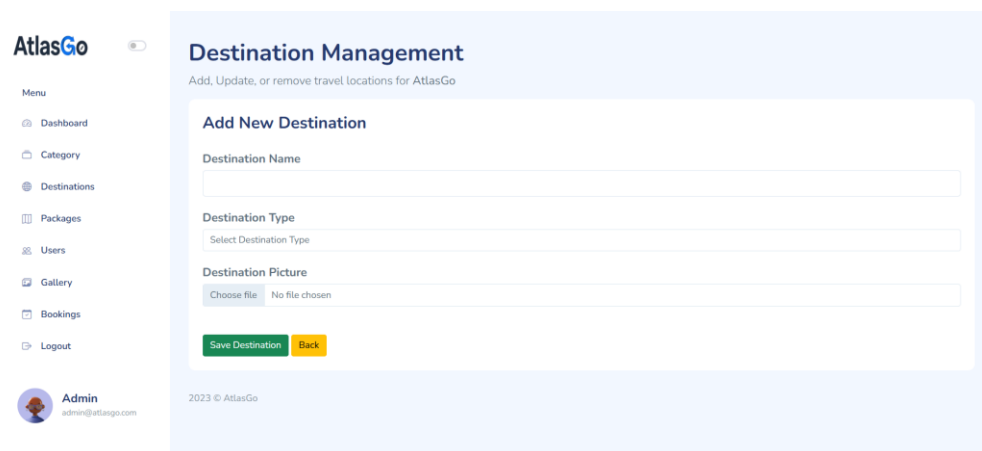


Figure 9.2.4: Admin Add Destination Form with Image Upload

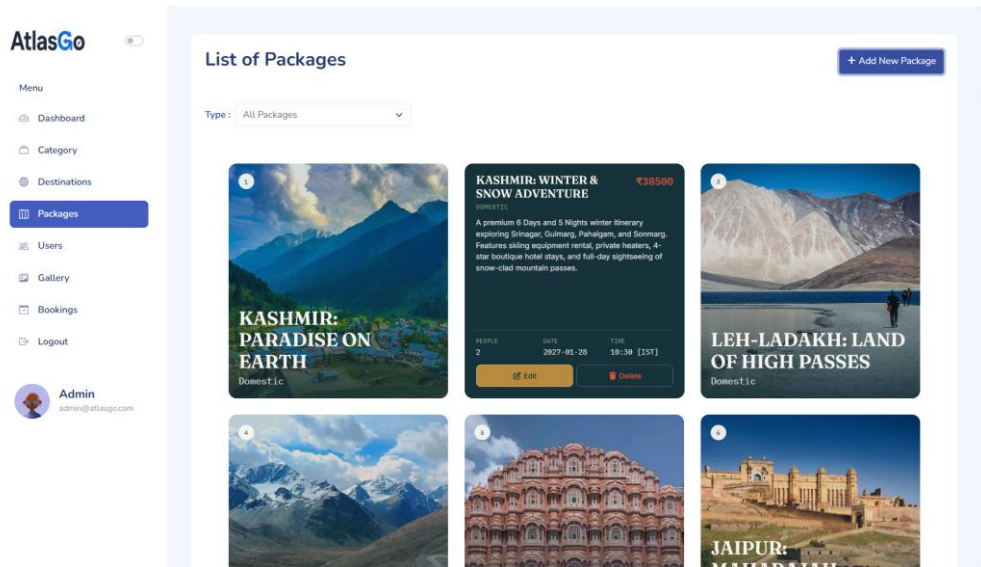


Figure 9.2.5: Admin Tour Package Inventory Table (Show Packages)

Figure 9.2.6: Admin Add Package Form with Price, Itinerary & Category

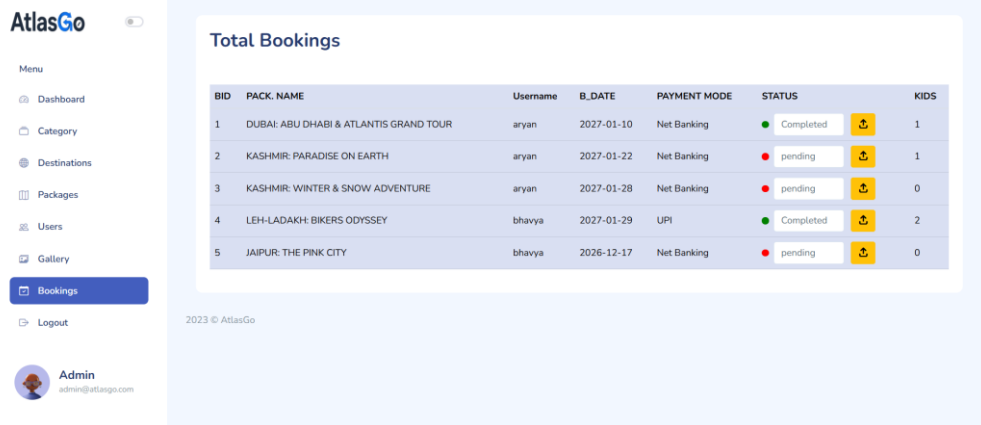


Figure 9.2.7: Admin Booking Management & Status Approval Table

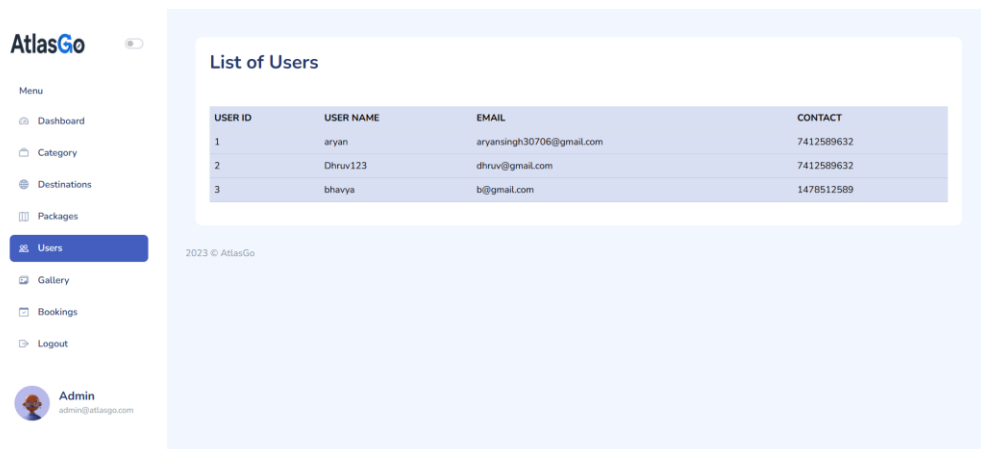


Figure 9.2.8: Admin User Management Directory (Show Users)

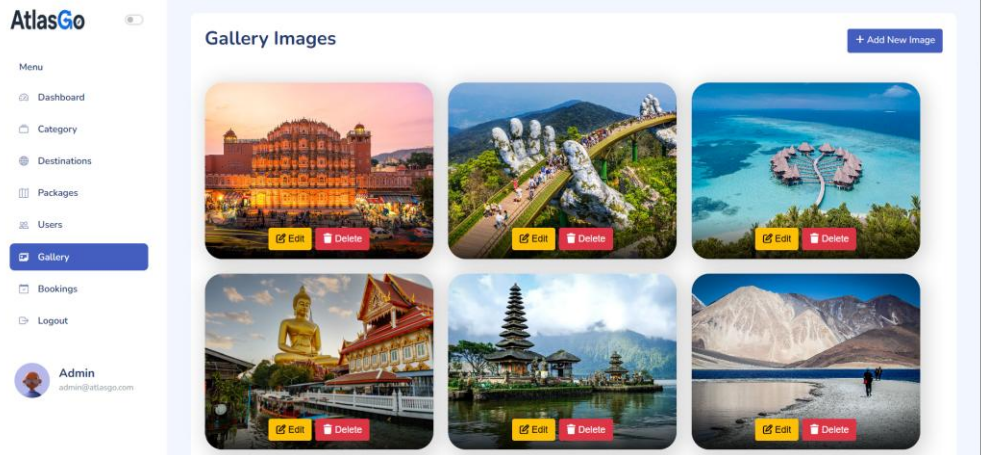


Figure 9.2.9: Admin Photo Gallery Showcase Grid (Show Gallery)

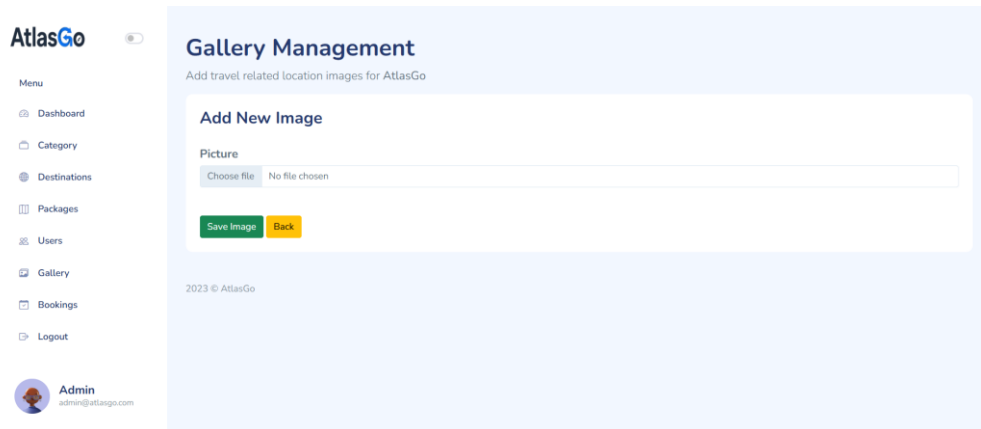


Figure 9.2.10: Admin Gallery Image Upload Form

10. ADVANTAGES OF THE SYSTEM

- Convenient Tour Browsing: Travelers can quickly filter and compare domestic and international tour packages from home.
- Dynamic Catalog Control: Administrators can add, edit, or delete destinations and packages dynamically without touching PHP code.
- Centralized Booking Tracking: Customers can track live reservation updates ('Pending', 'Processing', 'Completed') via their 'My Bookings' account.
- Responsive User Interface: Bootstrap 5 flexbox grid ensures seamless layout rendering across mobile phones, tablets, and desktops.
- Image File Automation: Automatic handling of image file uploads and server paths for package banners, destination thumbnails, and photo galleries.
- Secure Session Authentication: Protects user profile data and administrative dashboard routes using PHP sessions.

11. LIMITATIONS OF THE SYSTEM

- No Live Online Payment Gateway Integration: Currently records payment selections (UPI, Net Banking, Cash) as status fields rather than connecting to live payment APIs (e.g., Razorpay or Stripe).
- Lack of Automated Email / SMS Notifications: Booking confirmation receipts and status updates are displayed on-screen rather than emailed automatically to customers.
- Manual Seat Capacity Deductions: Bookings are logged, but automated deduction of available passenger seats per tour date is not currently enforced.
- Single Admin Role: The platform currently supports one primary administrator account rather than multi-tiered staff roles.

12. FUTURE ENHANCEMENTS

- Online Payment Gateway Integration: Integrate Razorpay or PayPal SDKs for real-time online transaction processing.
- Automated Email Invoicing: Incorporate PHPMailer to generate and email PDF booking receipts directly to customer inboxes.
- Customer Review & Rating Module: Allow verified travelers to submit ratings, reviews, and feedback for completed tours.
- Real-Time Inventory Seat Deduction: Automate available seat reductions upon booking submission.
- Wishlist & Package Comparison Tool: Enable customers to bookmark favorite packages and compare itineraries side-by-side.

13. CONCLUSION

The 'AtlasGo - Online Travel & Tour Booking System' successfully demonstrates the practical implementation of PHP and MySQL to create a functional, modern web application for the travel and tourism industry. The project covers key software engineering stages, including requirements analysis, database normalization, DFD and UML system modeling, dynamic web page development, session-based authentication, and administrative CRUD operations.

Through this project, we gained invaluable practical experience in server-side scripting, relational database design, responsive front-end development, and building real-world admin dashboards. AtlasGo satisfies all core functional goals and provides a solid, extensible architecture that can easily incorporate online payment gateways, automated email notifications, and customer review modules in future iterations.

14. REFERENCES

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